

Sahil Dinesh Zambre

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Education

Vellore Institute of Technology B.Tech in Computer Science and Engineering CGPA: 9.38	Aug. 2023 – Present Vellore, Tamil Nadu
MP Junior College Maharashtra State Board (HSC) Percentage: 87.17%	Mar. 2021 – June 2023 Pune, Maharashtra
Mar Ivanios Convent High School Maharashtra State Board (SSC) Percentage: 90.60%	Mar. 2010 – Mar. 2021 Pune, Maharashtra

Experience

Indian Institute of Technology, Roorkee <i>Research Intern</i> - Obtain hands-on insight into AI and ML, explore anomaly detection and dynamic bipartite graphs, and understand how theoretical concepts translate into practical problem-solving. - Strengthened research skills by reviewing 20+ academic papers, applying structured problem-solving, and translating AI theory into practical applications.	Dec. 2024 – Jan. 2025
Vice Chairperson GeoSpatial Club, VIT Vellore - Boosted online visibility with a 150% increase in social media followers and organized technical workshops fostering industry collaboration. - Secured sponsorships and partnerships 20% above fundraising targets, ensuring successful event execution.	Mar. 2025 – Mar. 2026

Project

Home Energy Tracker <i>Spring Boot, Kafka, MySQL, InfluxDB, Docker, Keycloak, Prometheus, Grafana</i> - Architected a distributed home energy monitoring platform using Spring Boot microservices, processing real-time electricity usage through Apache Kafka and InfluxDB time-series storage. - Built an event-driven pipeline for energy ingestion, usage aggregation, threshold-based alert generation, and email notifications using Kafka producers and consumers. - Implemented production-grade observability with Spring Boot Actuator, Micrometer, Prometheus, and Grafana to monitor application health, latency, and system metrics.	GitHub
Ride Booking Microservices Platform (Backend System) <i>Spring Boot, Kafka, Redis, MySQL, JPA, Eureka</i> - Architected and developed a distributed ride-booking platform using Spring Boot microservices with Netflix Eureka for service discovery and secure REST APIs. - Implemented JWT-based authentication and role-based authorization using Spring Security, ensuring secure communication across microservices. - Built an event-driven system using Apache Kafka and STOMP WebSockets to enable asynchronous ride processing and real-time ride status updates. - Optimized driver allocation using Redis Geospatial queries and incorporated Flyway database migrations, DTO-based architecture, scheduled jobs, and Retrofit for efficient inter-service communication.	GitHub
AI-Powered Email Assistant (Chrome Extension) <i>Spring Boot, WebFlux, Google Gemini API, React.js</i> - Developed a Chrome extension that dynamically generates AI-powered email replies with customizable tones (friendly, professional, formal) using Google Gemini API. - Implemented real-time DOM and URL monitoring with two Mutation Observers for seamless Gmail integration and dynamic UI injection. - Built a Spring Boot backend with WebFlux for asynchronous AI response fetching, ensuring low-latency performance. - Designed a responsive React.js frontend with Material-UI and Axios for smooth API communication, enhancing productivity and user experience.	GitHub
CryptoVerse – Cryptocurrency Dashboard <i>React.js, Redux Toolkit, Tailwind CSS, Ant Design, Chart.js</i> - Developed a responsive cryptocurrency analytics dashboard tracking 100+ cryptocurrencies using CoinRanking and News APIs, with interactive price trends using Chart.js. - Built a modern React application using Redux Toolkit, Ant Design, and Tailwind CSS, and deployed it on Netlify with GitHub-based version control.	Live GitHub

Technical Skills

Programming Languages: Java, C++, C, Python
CS Fundamentals: Object-Oriented Programming, Operating Systems, Computer Networks, DBMS
Web Technologies: Spring Boot, REST APIs, Microservices, WebSockets, React.js, Redux Toolkit, Tailwind CSS, HTML
Databases & SQL: MySQL, Redis, Hibernate, MongoDB
Dev Tools: Git, GitHub, Docker, Postman, Vercel, Render

Certifications and Achievements

- IBM** Advanced Generative AI Certification.
- AWS Educate** – Introduction to Generative AI Training Badge.
- Tata Group** – Data Visualization Virtual Job Simulation (Forage).
- Solved 800+ DSA problems across multiple coding platforms, including 500+ on LeetCode (Contest Rating: 1500+).
- Open Source Contributor, GirlScript Summer of Code (GSSoC).